

PROBLEM STATEMENT #43 | Developer Tools & IT Operations

Feature Flag & Dynamic Config Manager

1. Problem Context & Overview

A centralized configuration and feature toggling platform supporting percentage-based user cohort rollouts, environment overrides, and real-time client SDK flag synchronization.

Target Stakeholders / Actors: *Software Engineer, Release Manager*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall evaluate user targeting rules (e.g., User ID hash, Beta cohort) and return accurate boolean feature flag states.

Sample Acceptance Criteria: Pass: Target 10% rollout cohort receives enabled flag; Fail: User outside cohort receives enabled flag.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The feature flag evaluation API must execute in under 15 ms to minimize overhead when integrated into client critical paths.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.